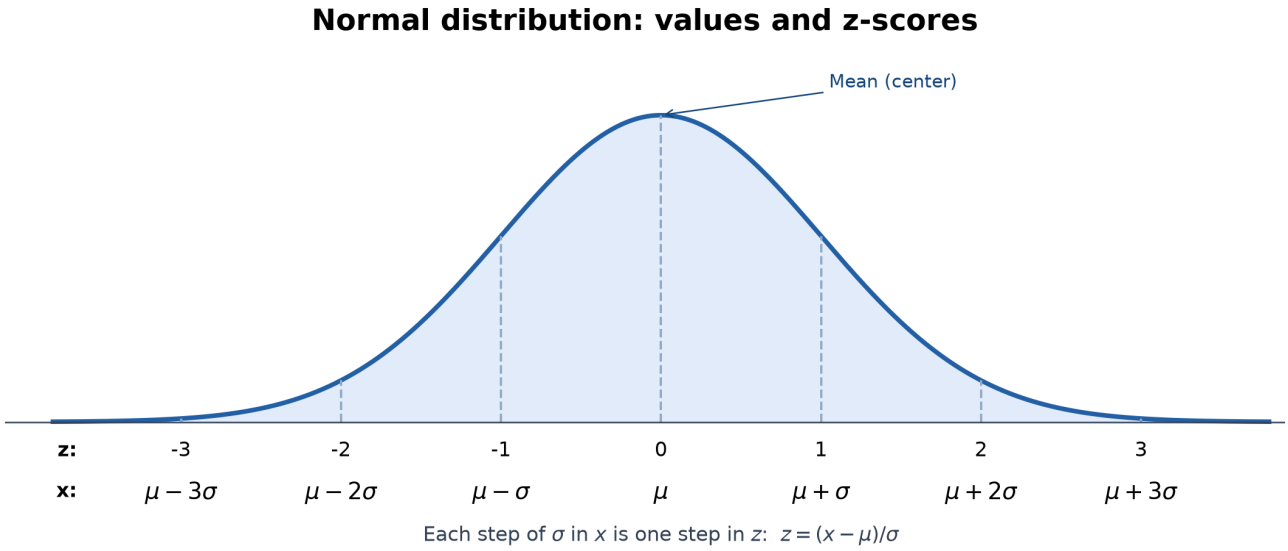
Lesson 8.1 - Standard Normal Distributions

> **Video:** Standard Normal Distribution Tables, Z Scores, Probability & Empirical Rule by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

The normal distribution

- A **bell-shaped**, symmetric curve, written $X \sim N(\mu, \sigma)$: the first number is the **mean**, the second the **standard deviation**.
- The mean sits at the center. Each standard deviation away from it is one unit of **z-score**.



Key formulas

Purpose	Formula
z-score of a value x	$z = \dfrac{x - \mu}{\sigma}$
Value x from a z-score	$x = \mu + z\sigma$
PDF (for reference)	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

You don't need the PDF to find probabilities in an intro course. Use z-tables (or calculus, see Lesson 8.2).

- **Positive** z → the value is **above** the mean. **Negative** z → **below** the mean.

The empirical rule (68-95-99.7)

Within	Percent of the data
$\mu \pm 1\sigma$	68%
$\mu \pm 2\sigma$	95%
$\mu \pm 3\sigma$	99.7%

- $68/2 = 34\%$
- $(95 - 68)/2 = 13.5\%$
- $(99.7 - 95)/2 = 2.35\%$
- $(100 - 99.7)/2 = 0.15\%$

The total area under the curve is 1 (100%).

Example 1 - $X \sim N(50, 10)$

(a) $\mu = 50$, $\sigma = 10$.

(b) Which x has $z = 1.4$?

$$x = \mu + z\sigma = 50 + 1.4(10) = 64$$

(c) What is the z-score of $x = 30$?

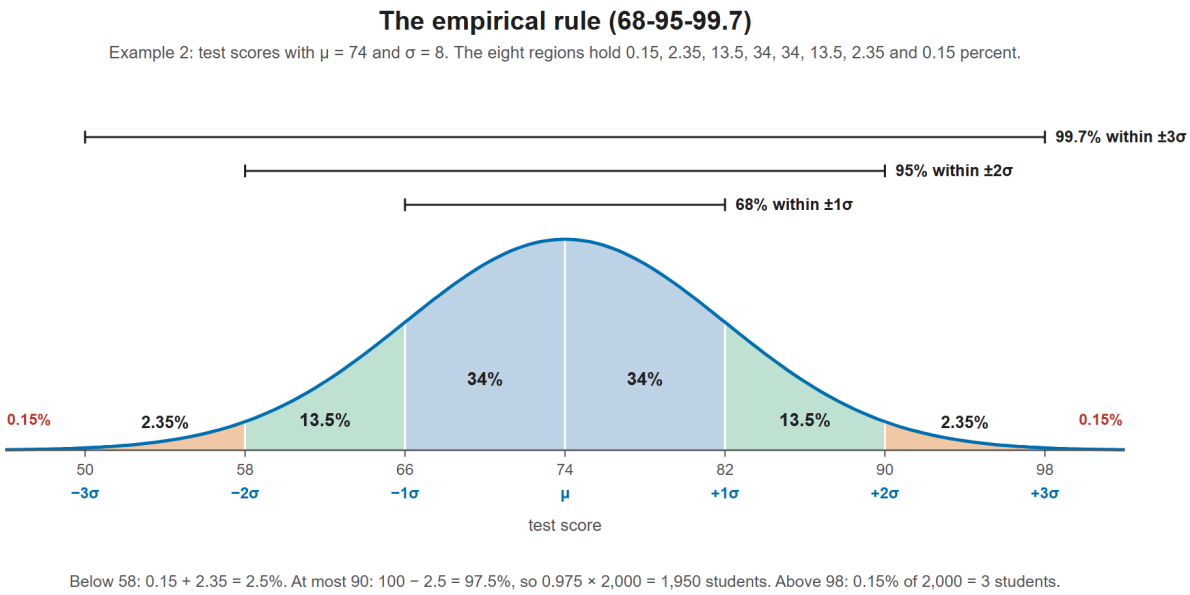
$$z = \frac{x - \mu}{\sigma} = \frac{30 - 50}{10} = -2$$

(d) Negative z-scores (like −2 for 30) are left of the mean. Positive ones (like 1.4 for 64) are right of it.

Example 2 - empirical rule: test scores, $\mu = 74$, $\sigma = 8$, 2,000 students

Mark the axis: 50, 58, 66, **74**, 82, 90, 98 (from -3σ to $+3\sigma$).

Question	Regions	Answer
(a) % below 58 ($< \mu - 2\sigma$)	0.15 + 2.35	**2.5%**
(b) P(66 < X < 82) ($\mu \pm 1\sigma$)	34 + 34	**68%**
(c) **How many** scored at most 90?	everything below $+2\sigma$ = 100 − (2.35 + 0.15) = 97.5%	$0.975 \times 2000 =$ **1,950 students**
(d) % at least 66	34 + 34 + 13.5 + 2.35 + 0.15	**84%**
(e) **How many** scored more than 98?	0.15%	$0.0015 \times 2000 =$ **3 students**



Below 58: 0.15 + 2.35 = 2.5%. At most 90: 100 − 2.5 = 97.5%, so 0.975 × 2,000 = 1,950 students. Above 98: 0.15% of 2,000 = 3 students.

The empirical rule on the test-score example (μ = 74, σ = 8). Each question in Example 2 adds up the regions it covers.

Using z-tables

When z is not a whole number, such as $z = 1.56$, the empirical rule can't be used. Use a **standard normal (z) table** instead.

A z-table gives the area to the LEFT of z.

- Find the row for the first decimal (**1.5**) and the column for the second decimal (**0.06**): area = **0.94062**.
- For $z = -0.43$: row **−0.4**, column **0.03**, area = **0.33360** (33.36%).

To find	Use
$P(X < x)$	the table value directly
$P(X > x)$	$1 -$ table value
$P(a < X < b)$	table(z_b) − table(z_a)

Example 3 - IQ scores, $\mu = 100$, $\sigma = 15$

(a) P(IQ < 80)

$$z = \frac{x - \mu}{\sigma} = \frac{80 - 100}{15} = -1.33 \;\Rightarrow\; \text{area} = 0.09176 \approx 9.18\%$$

(b) P(IQ > 136)

$$z = \frac{x - \mu}{\sigma} = \frac{136 - 100}{15} = 2.40 \;\Rightarrow\; \text{left area} = 0.99180$$

$$P(X > 136) = 1 - P(X < 136) = 1 - 0.99180 = 0.0082 = 0.82\%$$

(The video says "eight point two" at one point; the correct value is 0.0082, which is **0.82%**.)

(c) P(95 < IQ < 110)

$$z = \frac{x-\mu}{\sigma}: \qquad z_{110} = \frac{110-100}{15} = \frac{10}{15} \approx 0.67 \to 0.74857, \qquad z_{95} = \frac{95-100}{15} = \frac{-5}{15} \approx -0.33 \to 0.37070$$

$$P(95 < X < 110) = P(z < 0.67) - P(z < -0.33) = 0.74857 - 0.37070 = 0.37787 \approx 37.8\%$$

(d) IQ at the 90th percentile (work backwards from the area)

The left area is 0.90. In the table, $z = 1.28$ gives 0.89973 and $z = 1.29$ gives 0.90147. 0.89973 is closer to 0.90, so take $z = 1.28$.

$$x = \mu + z\sigma = 100 + 1.28(15) = 119.2$$

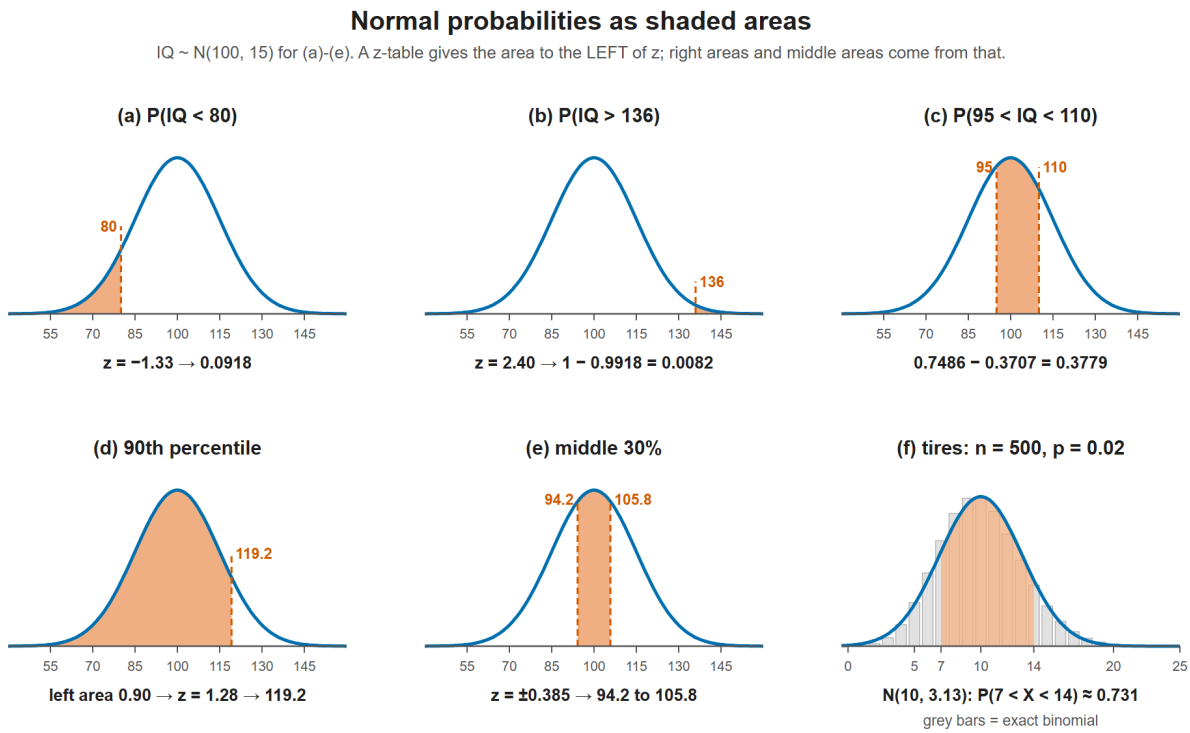
(e) Between which two IQs are the middle 30%?

The middle 30% leaves 15% on each side of the mean, so it runs from the **35th** to the **65th** percentile.

- For an area of 0.65: $z = 0.38$ gives 0.64803 and $z = 0.39$ gives 0.65173. 0.65 is about halfway, so $z \approx 0.385$.
- By symmetry, the 35th percentile is $z \approx -0.385$.

$$x = \mu + z\sigma: \qquad x_2 = 100 + 0.385(15) \approx 105.8, \qquad x_1 = 100 - 0.385(15) \approx 94.2$$

The middle 30% of IQs lie between about **94.2 and 105.8**.



Each IQ question as a shaded area. A left area comes straight from the z-table, a right area is 1 minus it, and a middle area is the difference of two left areas. For percentiles, start from the area and work back to x. In (f) the normal curve closely follows the exact binomial bars.

Example 4 - normal approximation to a binomial (defective tires)

2% of tires are defective, and a sample of $n = 500$ is checked. So $p = 0.02$, $q = 0.98$.

(a) Mean and standard deviation

$$\mu = np = 500(0.02) = 10, \qquad \sigma = \sqrt{npq} = \sqrt{500(0.02)(0.98)} = \sqrt{9.8} \approx 3.13$$

(b) P(fewer than 8 defective)

$$z = \frac{x - \mu}{\sigma} = \frac{8 - 10}{3.13} \approx -0.64 \;\Rightarrow\; P \approx 0.2611 = 26.1\%$$

(c) P(more than 15 defective)

$$z = \frac{x - \mu}{\sigma} = \frac{15 - 10}{3.13} \approx 1.60 \;\Rightarrow\; \text{left} = 0.9452,\; P = 1 - 0.9452 = 0.0548 = 5.48\%$$

(d) P(between 7 and 14 defective)

$$z = \frac{x - \mu}{\sigma}: \qquad z_{14} = \frac{14 - 10}{3.13} \approx 1.28 \to 0.89973, \qquad z_7 = \frac{7 - 10}{3.13} \approx -0.96 \to 0.16853$$

$$P(7 < X < 14) = P(z < 1.28) - P(z < -0.96) = 0.89973 - 0.16853 = 0.7312 = 73.1\%$$

> **Note:** the video uses the normal curve directly on the counts. Many textbooks add a **continuity correction** of ±0.5 (e.g. "fewer than 8" → $x < 7.5$) when approximating a discrete binomial with the continuous normal, which gives slightly different, more accurate answers.

Key takeaways

- $z = (x - \mu)/\sigma$ and $x = \mu + z\sigma$.
- Empirical rule: 68 / 95 / 99.7, which splits into 34 / 13.5 / 2.35 / 0.15 per region.
- A z-table gives the **left** area. Right area = 1 − left. Between = difference of two left areas.
- For percentiles, look up the **area** in the table to find z, then convert to x.